

# SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security  
COURSE CODE: CSA51

## COURSE OUTCOME COVERED

**CO1:** Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

## APPLICATION- BASED QUESTIONS

**Code of Conduct:**

I K. Lahari [192325118] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Lahari

Handwritten signature and a large circular stamp containing the number 50.

## APPLICATION- BASED QUESTIONS

**1.                      Scenario:                      Event                      Communication                      Security**

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

**2.                      Scenario:                      Intercepted                      Suspicious                      Message**

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

**3.                      Scenario:                      Secure                      Messaging                      System**

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

**4.                      Scenario:                      Suspicious                      Digital                      Asset                      Investigation**

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

**5.                      Scenario:                      Legacy                      Encryption                      in                      Corporate                      Communication**

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".



- Determine the most probable original message by reversing the encryption process.
- Critically evaluate why such a method is vulnerable in modern communication systems.
- Propose one practical enhancement to improve the confidentiality of messages in this scenario.

#### RUBRICS FOR CALCULATION:

| Criteria                  | Weightage | Level 4 – Exemplary                                                                           | Level 3 – Proficient                                        | Level 2 – Developing                                | Level 1 – Beginning                              |
|---------------------------|-----------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Understanding of Concepts | 25        | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design       | 30        | Well-planned, systematic implementation with optimal solution                                 | Correct implementation with minor issues                    | Basic implementation with errors or inefficiencies  | Incorrect or incomplete implementation           |
| Use of Tools              | 20        | Effective and advanced use of tools/technologies with proper configuration                    | Appropriate use of tools with correct execution             | Limited or inconsistent use of tools                | Incorrect or minimal use of tools                |

|                     |    |                                                                            |                                                |                                          |                                       |
|---------------------|----|----------------------------------------------------------------------------|------------------------------------------------|------------------------------------------|---------------------------------------|
| Analysis of Results | 15 | Insightful analysis with accurate interpretation and validation of results | Good analysis with reasonable interpretation   | Basic analysis with limited insights     | Poor or incorrect analysis of results |
| Documentation       | 10 | Clear, well-structured documentation with diagrams, code, and results      | Organized documentation with necessary details | Basic documentation with missing details | Poor or incomplete documentation      |



## Scenario : Event Communication Security (Caesar Cipher)

a. Encryption Process :-

Plain Text:- MEET AT MOON

Shift Value=4

Encryption formula  $\Rightarrow C = (P+K) \bmod 26$

Where

C - Cipher letter

P - plain letter

K - Key (or) Shift Value

|    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|
| A  | B  | C  | D  | E  | F  | G  | H  | I  | J  | K  | L  | M  |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 |
| N  | O  | P  | Q  | R  | S  | T  | U  | V  | W  | X  | Y  | Z  |
| 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

MEET AT MOON

$$M = (12+4) \bmod 26 \Rightarrow 16 \bmod 26 = 16 \Rightarrow Q$$

$$E = (4+4) \bmod 26 \Rightarrow 8 \bmod 26 = 8 \Rightarrow I$$

$$E = (4+4) \bmod 26 \Rightarrow 8 \bmod 26 = 8 \Rightarrow I$$

$$T = (19+4) \bmod 26 \Rightarrow 23 \bmod 26 = 23 \Rightarrow X$$

$$A = (0+4) \bmod 26 \Rightarrow 4 \bmod 26 = 4 \Rightarrow E$$

$$T = (19+4) \bmod 26 \Rightarrow 23 \bmod 26 = 23 \Rightarrow X$$

$$M = (12+4) \bmod 26 \Rightarrow 16 \bmod 26 = 16 \Rightarrow Q$$

$$O = (14+4) \bmod 26 \Rightarrow 18 \bmod 26 = 18 \Rightarrow S$$

$$O = (14+4) \bmod 26 \Rightarrow 18 \bmod 26 = 18 \Rightarrow S$$

$$N = (13+4) \bmod 26 \Rightarrow 17 \bmod 26 = 17 \Rightarrow R$$



∴ cipher text = QIIX EX RSSR

b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

The receiver knows that the message encrypted using a shift 4. They simply shift every ciphertext letter 4 positions backward.

Example:-

Q → M

I → E

X → T

R → N

After shifting all letters back, the original message becomes:

MEET AT NOON

2. Scenario: Intercepted Suspicious Message (Atbash cipher)

(a) Deduce the most likely original message.

Cipher Text: GSRH RH Z HVXIVG

The message is encrypted using Atbash cipher.

The Atbash cipher replaces every letter with its opposite letter the alphabet.

original

|   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | D | E | F | G | H | I | J | K | L | M |
| Z | Y | X | W | V | U | T | S | R | Q | P | O | N |

cipher



|          |   |   |   |   |   |   |   |   |   |   |   |   |   |
|----------|---|---|---|---|---|---|---|---|---|---|---|---|---|
| original | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| cipher   | M | L | K | J | I | H | G | F | E | D | C | B | A |

By Applying this rule

G → T

S → H

R → I

H → S

R → I

H → S

Z → A

H → S

V → E

X → C

I → R

V → E

G → T

∴ ORIGINAL MESSAGE ⇒ "THIS IS A SECRET"

(b) Justification

- \* It uses a fixed Substitution rule.
- \* Every alphabet letter maps to exactly one opposite letter.
- \* Repeating Encrypted words produce repeating decrypted words.
- \* The single-letter word Z becomes A, matching normal english.

Characteristics :-

- \* Monoalphabetic Substitution cipher.
- \* Uses a fixed reverse alphabet.
- \* Easy to encrypt and decrypt.



3.

### Scenario : Secure Messaging System (Vigenere Cipher)

Plain text : HELLO

Keyword : KEY

Step 1 : Repeat the Keyword

Since the message have five words letters, the Keyword becomes

|            |   |   |   |   |   |
|------------|---|---|---|---|---|
| Plain text | H | E | L | L | O |
| Keyword    | K | E | Y | K | E |

Step 2 : Convert letter into numbers

|        |   |   |    |    |    |   |    |
|--------|---|---|----|----|----|---|----|
| Letter | H | E | L  | O  | K  | E | Y  |
| Value  | 7 | 4 | 11 | 14 | 10 | 4 | 24 |

Step 3 : Apply formula

Encryption formula : Cipher = (Plain + Key) mod 26

| Plain  | Key    | $(P+K) \bmod 26$        | Cipher |
|--------|--------|-------------------------|--------|
| H (7)  | K (10) | $(7+10) \bmod 26 = 17$  | R      |
| E (4)  | E (4)  | $(4+4) \bmod 26 = 8$    | I      |
| L (11) | Y (24) | $(11+24) \bmod 26 = 9$  | J      |
| L (11) | K (10) | $(11+10) \bmod 26 = 21$ | V      |
| O (14) | E (4)  | $(14+4) \bmod 26 = 18$  | S      |



∴ Cipher Text :- RIJVS

b. Explain how the repeating Keyword influences each character of the ciphertext.

- \* The Keyword KEY repeats as KEYKE.
- \* Each Keyword letter determines the shift for one Plaintext letter.
- \* Different letters are shifted by different amounts.
- \* This makes the ciphertext more secure than Caesar cipher because the shift is not constant.

#### 4. Scenario: Suspicious Digital Asset Investigation (Steganography)

Introduction:-

- \* Steganography is the process by which secret information is hidden inside digital files such as images, videos, audio files or documents.
- \* Unlike encryption, which hides the content, steganography hides the existence of communication itself.

##### a. Detecting Hidden Data:-

Investigators perform several analyses.

##### 1. Least Significant Bit (LSB) Analysis:-

Most hidden messages modify the least significant bits of image pixels.



2. Histogram Analysis :-

Hidden data changes the statistical distribution of pixel values.

3. Metadata Inspection :-

Checks whether unusual metadata or the hidden attributes exists.

File Size Comparison :- If an image is unusually large without visible changes, hidden data may exist.

Statistical Analysis :-

Detects abnormal pixel patterns.

Steganalysis Tools :-

- \* OpenStego
- \* Steg detect
- \* Steg Expose
- \* Steg Secret

(b) Techniques to improve Security :-

\* Encrypt Before embedding :- Encrypt the confidential message using AES before hiding it.

\* Random pixel embedding :- Hide data in randomly selected pixels using a Secret Key. This prevents attackers from locating hidden information easily.



scenario :- Legacy encryption incorporate communication  
(Rail Fence cipher)

Introduction:-

\* The rail fence cipher is a transposition cipher that encrypts messages by rearranging the letters in a zig-zag pattern

Given Cipher Text:-

WECRLTEE RDSOEEFEADCAIVDEN

(a) Determine the Original message.

Using the 3-rail fence cipher, the decrypted message is

Original Message:-

WE ARE DISCOVERED FLEE AT ONCE

Working Principle

Encryption:-

The message is written diagonally across the three rails.

Ex:-

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| W | . | . | . | . | E | . | . | . | C | . | . | . | . |
| . | E | . | . | R | . | D | . | S | . | O | . | . | . |
| . | . | A | . | . | . | I | . | . | . | . | . | . | . |

The rows are then read one after another to produce the ciphertext.



## Decryption:-

The receiver reconstructs the zigzag pattern and reads the letters diagonally to recover the original message.

### b, Why it is Vulnerable:-

The rail fence cipher is considered insecure because.

- \* It only rearrange letters.
- \* Letter frequencies remain unchanged.
- \* Easy to guess number of rails.
- \* Vulnerable to brute-force attacks.
- \* Modern Computers decrypt it within seconds
- \* Provides no authentication.
- \* Does not ensure message integrity.

### c, propose one practical enhancement to improve the confidentiality of messages in scenario.

Recommended enhancements:

- \* Use AES (Advanced Encryption Standard) for strong encryption before transmission.
- \* Alternatively, combine transposition with a strong Substitution Cipher to increase security.